



Does the Type of Visualization Influence the Mode of Cognitive Control in a Dynamic System?

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Abstract. This study investigates the influence of a visualization system on the mode of cognitive control adopted in a navigation task. It relies on a driving simulator experiment. It uses data clustering methods, which help identify three patterns of in-vehicle data corresponding to three different modes of control. The study shows that Augmented Reality HUD supports drivers, since it contributes to avoiding the adoption of a scrambled mode of control. However, it promotes an opportunistic mode of control characterised by a higher but limited anticipation.

Keywords: Cognitive control modes · Operator assistance · Visualization system · Augmented reality · Data clustering

1 Introduction

When interacting with a system in a dynamic environment, humans use internal models (a model of the system as well as a model of the dynamics of the environment) and external data. Cognitive control refers to functions that allow information processing and behaviour to vary adaptively rather than remaining rigid and inflexible and to rely in some cases, more on internal data and in other cases, more on external. Within the framework of cognitive system engineering, Hollnagel [1] identified four modes of cognitive control, from the most reactive (determined by the occurrence of external data and characterized by a short processing time) to the most proactive (relying on internal data processing and characterized by a longer processing time). The control mode changes from “scrambled” to “opportunistic” when there is an increase in the predictability of the situation or in the available time, and from “opportunistic” to “tactical” or “strategic” if the predictability of the situation and/or the available time continues to improve. It is assumed that cognition reliability is low for both scrambled and opportunistic modes, whereas it is assumed to be relatively high in tactical and

strategic modes [2]. One major question is whether a man-machine interface can influence the adoption of a specific control mode.

This paper investigates the influence of visualization on the cognitive control mode used by drivers in a navigation task. Navigation in an unknown road environment represents a costly cognitive activity for drivers. Numerous studies have shown the difficulties they encounter in planning and following routes [3]. Different navigation systems have been designed to facilitate this task. Amongst them are Augmented Reality-based Head Up-Displays which are expected to facilitate drivers' perception of useful information and to therefore increase the predictability of a complex situation [4], [5]. This paper aims at assessing the value of an Augmented Reality Head Up Display (AR HUD) in comparison to a classical windshield HUD (W-HUD).

2 Method

2.1 Participants and Equipment

A driving simulator experiment was carried out with 32 participants. It was conducted with the simulator of the Institute for Technological Research System X: the Dr SIHMI (Driving Simulator for Human Machine Interaction studies) platform. This static simulator consists of an automobile cockpit (including a steering wheel, the acceleration and brake pedals, a gear lever, an AR HUD or a classical W-HUD, and a standard GPS), and a 180° curved projection screen giving a simulation of different driving situations with the SCANer simulation software.

Figure 1a gives a view of the W-HUD, whereas Fig. 1b presents the AR HUD. The classical W-HUD displayed a map indicating the route to follow, reproducing the system used on BMW Series 7 vehicles. With a surface area of $9 \times 15^\circ$, the display unit of the AR HUD projects virtual information in the driving environment. In this case, arrows indicated the direction to follow; it appeared 400 m before the place where the manoeuvre is possible. The AR surface of the unit was $8 \times 15^\circ$. In both conditions, additional information appeared on a fixed section at the bottom of the windscreen: the speed limit, vehicle speed, next change of direction (shown through a “turn by turn” icon) and the remaining distance before reaching it, and the direction sign.

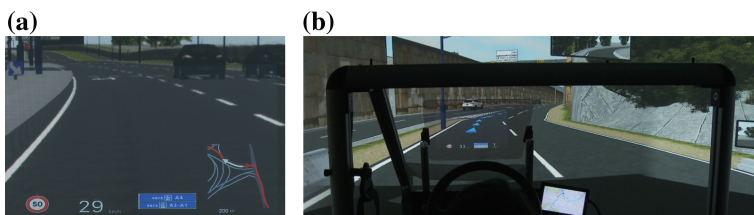


Fig. 1. a. W-HUD and direction sign. b. AR HUD and GPS Waze

2.2 Experiment Design and Procedure

The experiment sessions lasted 1 h30. Participants were first invited to read an information document and then sign consent forms. They were also introduced to the simulator, interfaces, and the test (5 min). Then, a learning phase consisted of driving with and without augmented reality. Afterwards, participants were required to drive in two complex geographical areas. In both areas, half the participants drove with the AR HUD and the other half with the W-HUD. Each participant tested both interfaces. The running order was counterbalanced. In case of failure, the driving phase was interrupted, and participants had to start over again. As participants could replay this phase up to three times (should they fail the first and second driving test), the duration of this phase varied between 20 and 30 min. A clarifying interview was held after each experimental phase. Finally, participants were asked to complete a questionnaire relating to their preferences and their profile (of about 10 min).

2.3 Data Collection

Assessment of interfaces is based upon objective data (e.g. vehicle speed, acceleration, actions on the brake pedal, the acceleration pedal and the steering wheel, Time To Collision or TTC, lane-change duration) and subjective data (such as a comfort score) which was collected during a clarifying interview.

2.4 Hypothesis

As in previous studies [6], we expected to identify behavior showing the drivers' anticipatory capacity and revealing three modes of cognitive control. The scrambled mode should be associated to rough actions on the pedals and steering wheel, short following distances and few mirror checks. Conversely, a tactical mode should lead to smooth actions, adequate following distances and frequent mirror checks. In the opportunistic control mode, drivers' actions are determined by the salient features of the situation; this control mode should lead to reduced speed and few mirror checks.

2.5 Analysis

A two-step multivariate segmentation analysis was carried out, where clustering techniques are used for both variables and participants. In the first step, subsets of strongly correlated variables were detected, and the ones that provide the same kind of information were placed into the same group. The resulting groups reveal the main dimensionalities of the data, and a representative variable was determined for each of them. These representatives were used in the second step to segment the participants via classical clustering algorithms. Finally, the interpretation of the clusters of participants was made by means of univariate tests and Classification and regression trees (CART).

3 Results

This paper presents preliminary results concerning one task (changing lane in an X-shaped interchange). Only 3 out 32 participants had to redo this task due to a failure. For these participants, only the data collected during the second run were analyzed.

Objective variables that measure vehicle's performance (speed, actions on the pedals or the steering wheel...) were used to segment the participants into groups whereas variables related to the interaction with other vehicles (TTC, rearview mirror consultation...) along with subjective variables were used to interpret the resulting clusters.

Figure 2 shows the dendrogram for the clustering of vehicle variables. It appears from the examination of the bootstrapped mean-adjusted rand index that the three-partition solution is the most stable.

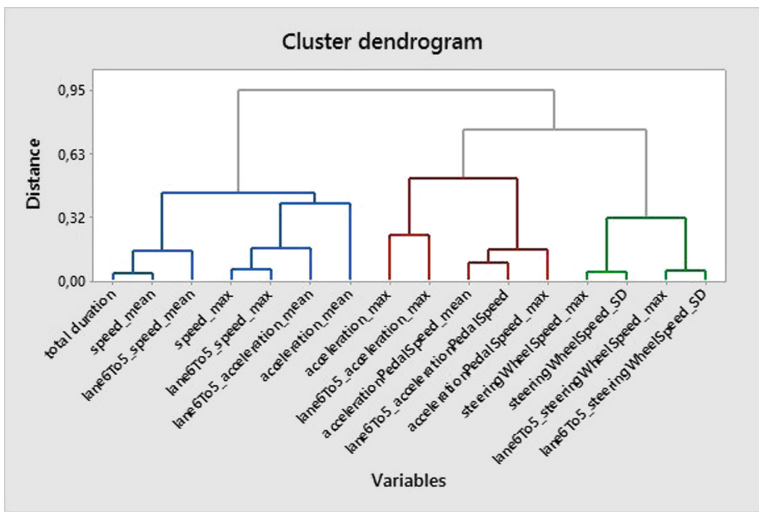


Fig. 2. Variables dendrogram

Examination of the composition of the clusters of variables and their factor loadings (squared correlations between variables and the first principal component of their parent group), suggests that Cluster 1 (in blue) refers to the speed of the vehicles, Cluster 2 (in red) to the smoothness of the actions upon the acceleration pedal and Cluster 3 (in green) to the smoothness of the actions upon the steering wheel.

In each cluster, the variable with the greatest factor loading is chosen as the representative of the group: maximal speed for Cluster 1, maximal acceleration for Cluster 2 and standard deviation of the actions upon the steering wheel for Cluster 3. The subsequent clustering of participants in groups is based on these variables and it is performed via the hierarchical agglomerative criterion of Ward reinforced with kmeans partitioning.

Figure 3 displays the resulting clustering of participants: the first group (in green, $N = 10$) is composed of participants who didn't experience Augmented Reality or did not make use of it, the second group (in red, $N = 6$) is mainly composed of participants who experienced AR HUD, and the third group (in black, $N = 15$) is well balanced between AR HUD and W-HUD.

Univariate tests and CART algorithm were used to describe the main discriminating variables of the clusters of participants. It resulted in the comfort score reported by the participants in the third group being significantly higher than in the two others, $F(2, 30) = 5.14$, $p = 0.013$.

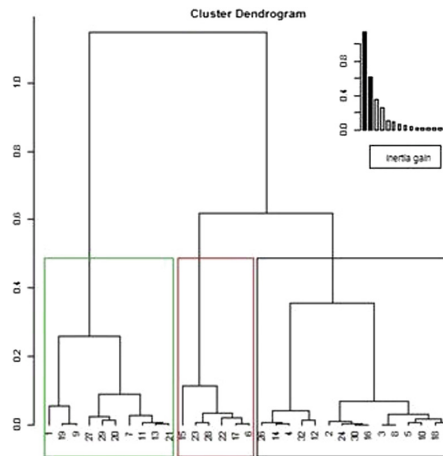


Fig. 3. Clusters of participants from the Ward dendrogram

In the first group, control actions are rougher with more variability than in the other groups (cf. Table 1). Furthermore, the task is performed more quickly and the TTC between the ego vehicle and the front left car is significantly lower.

Table 1. Main discriminating variables of Class 1

Class 1	v.test	Mean in class	Overall mean	sd in class	Overall sd	p.value
SteeringWheelSpeed_SD (rd.s^{-1})	4.49	0.28	0.21	0.04	0.06	<0.001
SteeringWheelSpeed_max (rd.s^{-1})	4.01	2.06	1.48	0.53	0.55	<0.01
Acceleration_max (m. s^{-2})	2.54	4.37	3.72	0.74	0.96	0.01

The second group is characterized by a lower maximal speed, more time to achieve the task, smoother longitudinal and lateral actions with more reduced variability (cf. Table 2) and, furthermore, by a later consultation of the rearview mirrors.

Table 2. Main discriminating variables of Class 2

Class 2	v.test	Mean in class	Overall mean	sd in class	Overall sd	p value
Total_duration (s)	3.74	16.74	13.81	2.72	2.10	<0.01
Speed_max (km/h)	-4.18	41.65	53.10	4.76	7.35	<0.01
Acceleration_max (m s^{-2})	-2.86	2.70	3.73	0.52	0.96	<0.01
SteeringWheelSpeed_max (rd s^{-1})	-2.42	0.99	1.48	0.14	0.55	0.01
SteeringWheelSpeed_SD (rd s^{-1})	-2.92	0.15	0.21	0.03	0.06	<0.01

4 Discussion and Conclusion

Clustering methods help identify three behavioural classes. With a significantly higher score of comfort, one class is distinguished from the others. It is composed of participants who were provided with either the AR-HUD or the W-HUD. It is worth noting that a large majority of them reported, during clarifying interviews, that they mainly used the color of the direction sign displayed on the bottom of HUD and matched it with the sign posts rather than using AR (blue arrows) or the W-HUD (map).

In the other classes, most participants reported a negative feeling. One of them is mostly composed of participants who used the W-HUD. They made abrupt actions and TTC was lower than the TTC observed in the other classes. These features are reminiscent of a scrambled mode. One class is mostly composed of participants who used the AR-HUD. Compared with other participants, they take longer to make the lane change, and they look at the mirrors later. They reduce their speed, in order to give themselves enough time to react. These features are reminiscent of an opportunistic mode of control.

In this study, benefits of AR HUD in terms of driver assistance prove therefore to be mixed. It supports drivers, since it contributes to avoiding the adoption of a scrambled mode of control. However, it promotes an opportunistic mode of control characterised by a higher, but limited anticipation. Referring to the model presented by Ross and Burnett [3], one may assume that AR HUD supports the stage of the navigation task consisting of pinpointing the location of the next maneuver within the road environment and identifying which direction to travel at the maneuver. It does not support, however, the former stage of “preview” consisting in building an early mental picture of the maneuver.

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